Weight and local gravity

antocciphysics | TEACHER KEY | CPA

Learning target: Distinguish mass from gravitational weight.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 5 course days later.*
$a=7$ m/s² right; $v_f=28$ m/s right. Constant net force and mass throughout.

02. Explain a worked example (worked)
12 N and 6 N downward toward each body's center; mass unchanged at 3 kg.

03. Look and explain (model)
Mass 2 kg in both places; weights 20 N and 10 N.

04. Build the solution (practice)
15 N; 6 kg weighs 30 N, twice as much.

05. Use the idea (reason)
No. Scale measures contact/support force; shared free fall can give near-zero reading while gravity acts.

06. Choose a method (mixed) *Review: Unit 1, Day 13; 12 course days later.*
Net force 24 N right; acceleration 6 m/s² right.

07. Exit check (exit)
4 kg and 12 N.

08. STRETCH: Apply it another way (stretch)
No: 2 kg at 6 N/kg and 3 kg at 4 N/kg both weigh 12 N.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 5 course days later.*
x interval [-55, -45] km; y interval [63, 77] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

02. Explain a worked example (worked)
12 N and 6 N downward toward each body's center; mass unchanged at 3 kg.

03. Represent and interpret (model)
Mass 2 kg in both places; weights 20 N and 10 N.

04. Solve and justify (practice)
588 N; 96 N; g=240/60=4 N/kg.

05. Analyze the reasoning (reason)
No. Scale measures contact/support force; shared free fall can give near-zero reading while gravity acts.

06. Choose a method (mixed) *Review: Unit 1, Day 13; 12 course days later.*
Net -32 N; a=-8 m/s^2. Need current velocity: leftward motion speeds up; rightward motion slows.

07. Exit check (exit)
Mass 5 kg; new weight 10 N.

08. STRETCH: Apply it another way (stretch)
Mass 6 kg; first g=12 N/kg. Weight changes with field, not necessarily mass.